```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving student-mat.csv to student-mat.csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("student-mat.csv", sep=';')
df.head()
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	health	abs
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	3	
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	3	
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	3	
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	5	
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	5	

5 rows × 33 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      395 non-null    object
 1   sex         395 non-null    object
 2   age         395 non-null    int64
 3   address     395 non-null    object
 4   famsize     395 non-null    object
 5   Pstatus     395 non-null    object
 6   Medu        395 non-null    int64
 7   Fedu        395 non-null    int64
 8   Mjob        395 non-null    object
 9   Fjob        395 non-null    object
 10  reason      395 non-null    object
 11  guardian    395 non-null    object
 12  traveltime  395 non-null    int64
 13  studytime   395 non-null    int64
 14  failures    395 non-null    int64
 15  schoolsup   395 non-null    object
 16  famsup      395 non-null    object
 17  paid        395 non-null    object
 18  activities  395 non-null    object
 19  nursery     395 non-null    object
 20  higher      395 non-null    object
 21  internet    395 non-null    object
 22  romantic    395 non-null    object
 23  famrel      395 non-null    int64
 24  freetime    395 non-null    int64
 25  goout       395 non-null    int64
 26  Dalc        395 non-null    int64
 27  Walc        395 non-null    int64
 28  health      395 non-null    int64
 29  absences    395 non-null    int64
 30  G1          395 non-null    int64
 31  G2          395 non-null    int64
 32  G3          395 non-null    int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB
```

```
df.isnull().sum()
```

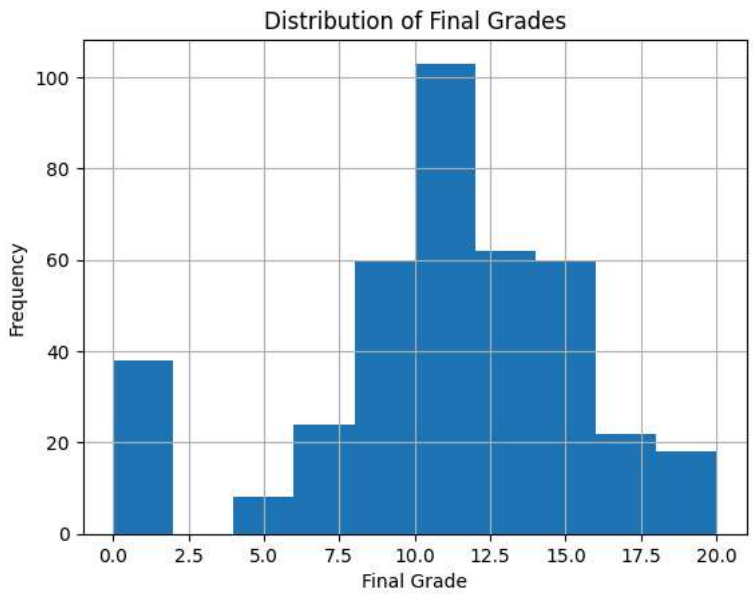
	0
school	0
sex	0
age	0
address	0
famsize	0
Pstatus	0
Medu	0
Fedu	0
Mjob	0
Fjob	0
reason	0
guardian	0
traveltime	0
studytime	0
failures	0
schoolsup	0
famsup	0
paid	0
activities	0
nursery	0
higher	0
internet	0
romantic	0
famrel	0
freetime	0
goout	0
Dalc	0
Walc	0
health	0
absences	0
G1	0
G2	0
G3	0

dtype: int64

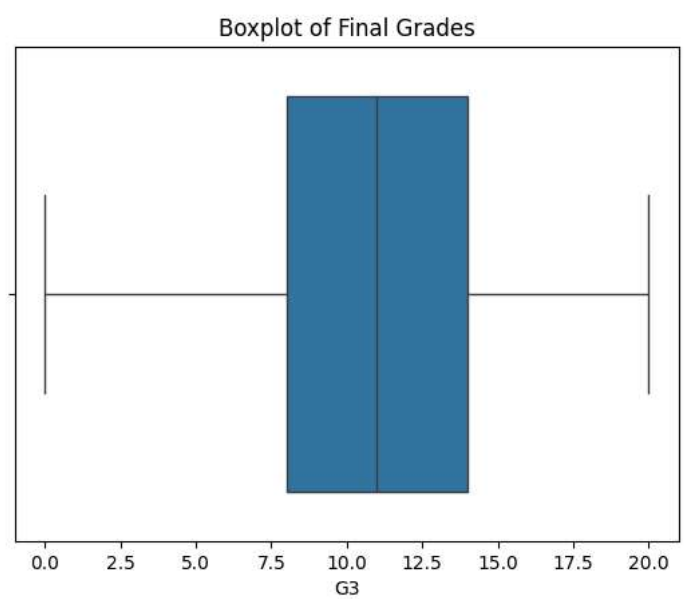
```
df.describe()
```

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	goout	Dalc	
count	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.00
mean	16.696203	2.749367	2.521519	1.448101	2.035443	0.334177	3.944304	3.235443	3.108861	1.481013	2.29
std	1.276043	1.094735	1.088201	0.697505	0.839240	0.743651	0.896659	0.998862	1.113278	0.890741	1.28
min	15.000000	0.000000	0.000000	1.000000	1.000000	0.000000	1.000000	1.000000	1.000000	1.000000	1.00
25%	16.000000	2.000000	2.000000	1.000000	1.000000	0.000000	4.000000	3.000000	2.000000	1.000000	1.00
50%	17.000000	3.000000	2.000000	1.000000	2.000000	0.000000	4.000000	3.000000	3.000000	1.000000	2.00
75%	18.000000	4.000000	3.000000	2.000000	2.000000	0.000000	5.000000	4.000000	4.000000	2.000000	3.00
max	22.000000	4.000000	4.000000	4.000000	4.000000	3.000000	5.000000	5.000000	5.000000	5.000000	5.00

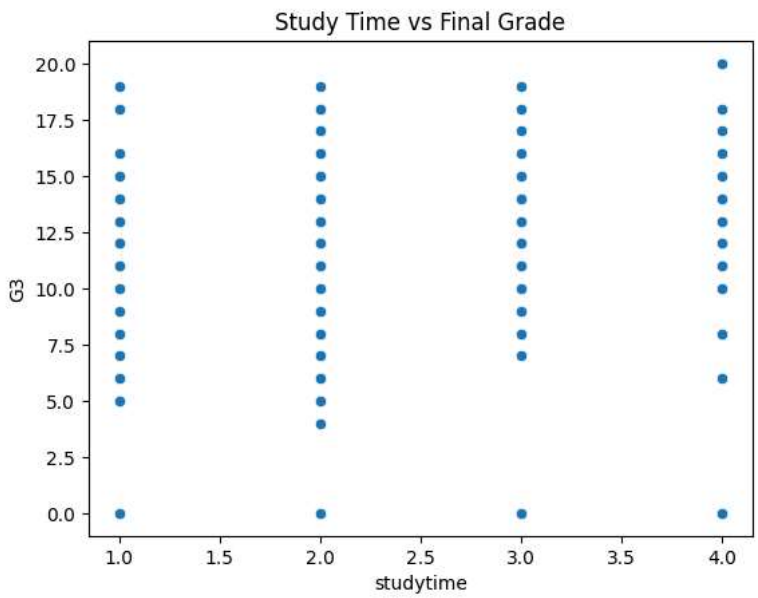
```
df['G3'].hist()
plt.title("Distribution of Final Grades")
plt.xlabel("Final Grade")
plt.ylabel("Frequency")
plt.show()
```



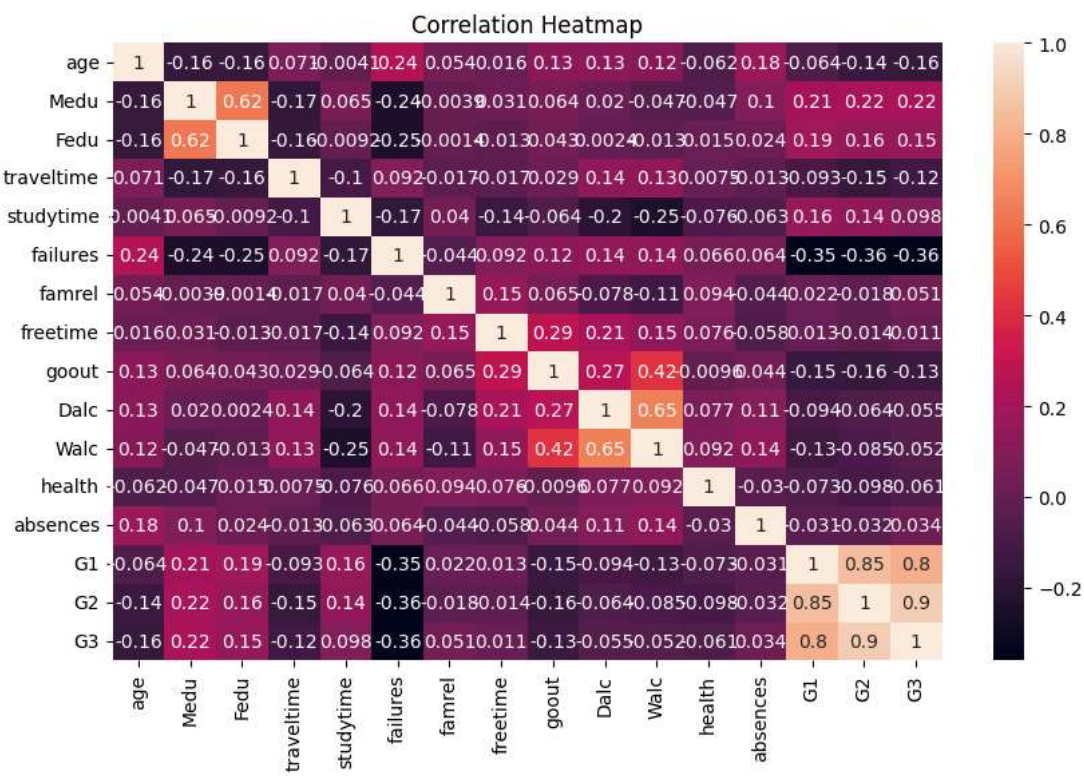

```
sns.boxplot(x=df['G3'])
plt.title("Boxplot of Final Grades")
plt.show()
```



```
sns.scatterplot(x='studytime', y='G3', data=df)
plt.title("Study Time vs Final Grade")
plt.show()
```

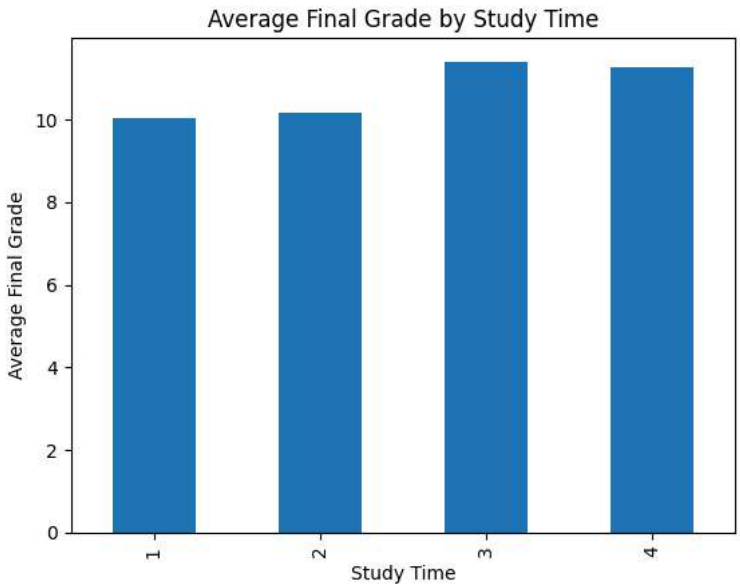



```
plt.figure(figsize=(10,6))

numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(numeric_df.corr(), annot=True)

plt.title("Correlation Heatmap")

plt.show()
```

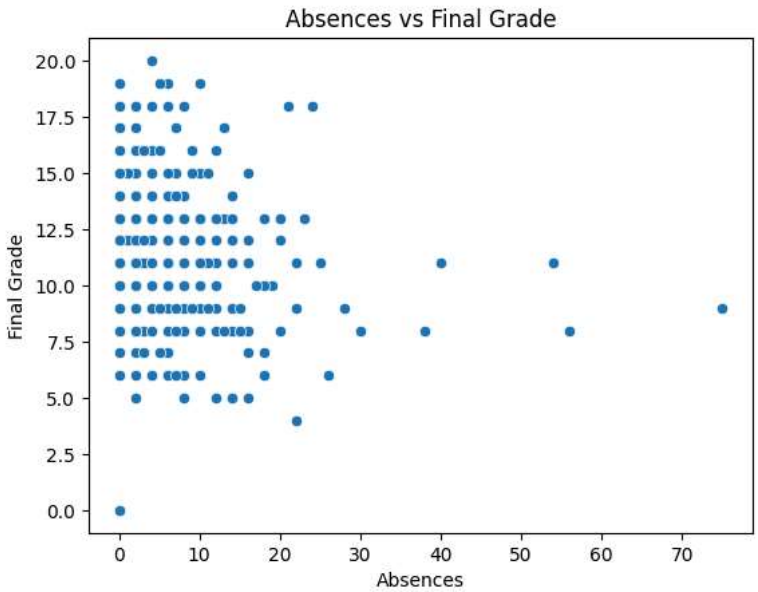



```
study_grade = df.groupby("studytime")["G3"].mean()

study_grade.plot(kind="bar")
```

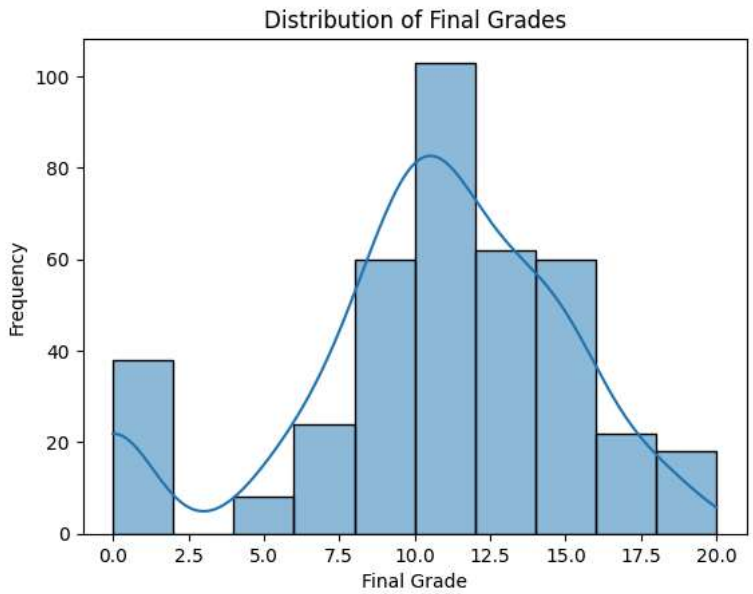
```
plt.title("Average Final Grade by Study Time")
plt.xlabel("Study Time")
plt.ylabel("Average Final Grade")
plt.show()
```




```
sns.scatterplot(x="absences", y="G3", data=df)

plt.title("Absences vs Final Grade")
plt.xlabel("Absences")
plt.ylabel("Final Grade")
plt.show()
```




```
sns.histplot(df["G3"], bins=10, kde=True)

plt.title("Distribution of Final Grades")
plt.xlabel("Final Grade")
plt.ylabel("Frequency")
plt.show()
```




```
df["pass"] = df["G3"].apply(lambda x: 1 if x >= 10 else 0)
```

```
numeric_df = df.select_dtypes(include=['number'])

X = numeric_df.drop(["G3","pass"], axis=1)
y = df["pass"]
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Logistic Regression Model

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

```
Accuracy: 0.9493670886075949
```

Random Forest Method

```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier()

rf.fit(X_train, y_train)

rf_pred = rf.predict(X_test)

rf_accuracy = accuracy_score(y_test, rf_pred)

print("Random Forest Accuracy:", rf_accuracy)
```

```
Random Forest Accuracy: 0.9113924050632911
```

```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, rf_pred)

sns.heatmap(cm, annot=True)

plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```